



POWERING
GREAT
MINDS



National Institute Of Business Management Diploma In Software Engineering

Computer Networks Course Work

Submitted by

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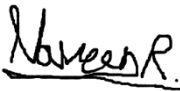
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Declaration

I, Naveen Rashmika Devinda, hereby declare that this course work report submitted for GADSE251F Computer network module in fourth trimester is an original work done by me under the guidance of Mr.Supun Asanga.also declare that this project record has not been submitted to any other institute.

Signature:


.....

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Introduction

For my computer network module course work i built a network for a call center named ConnectX, located in colombo 5. So in this report im going to attach all of my devices, diagrams, assumptions and all other details about my network.

Network Requirements

Ground floor,

- marketing dept for 30 users
- admin dept for 28 users
- server room for the it team
- reception
- customer waiting area with wifi

1st floor,

- US call center for 120 users
- Video conferencing room
- training center with wifi

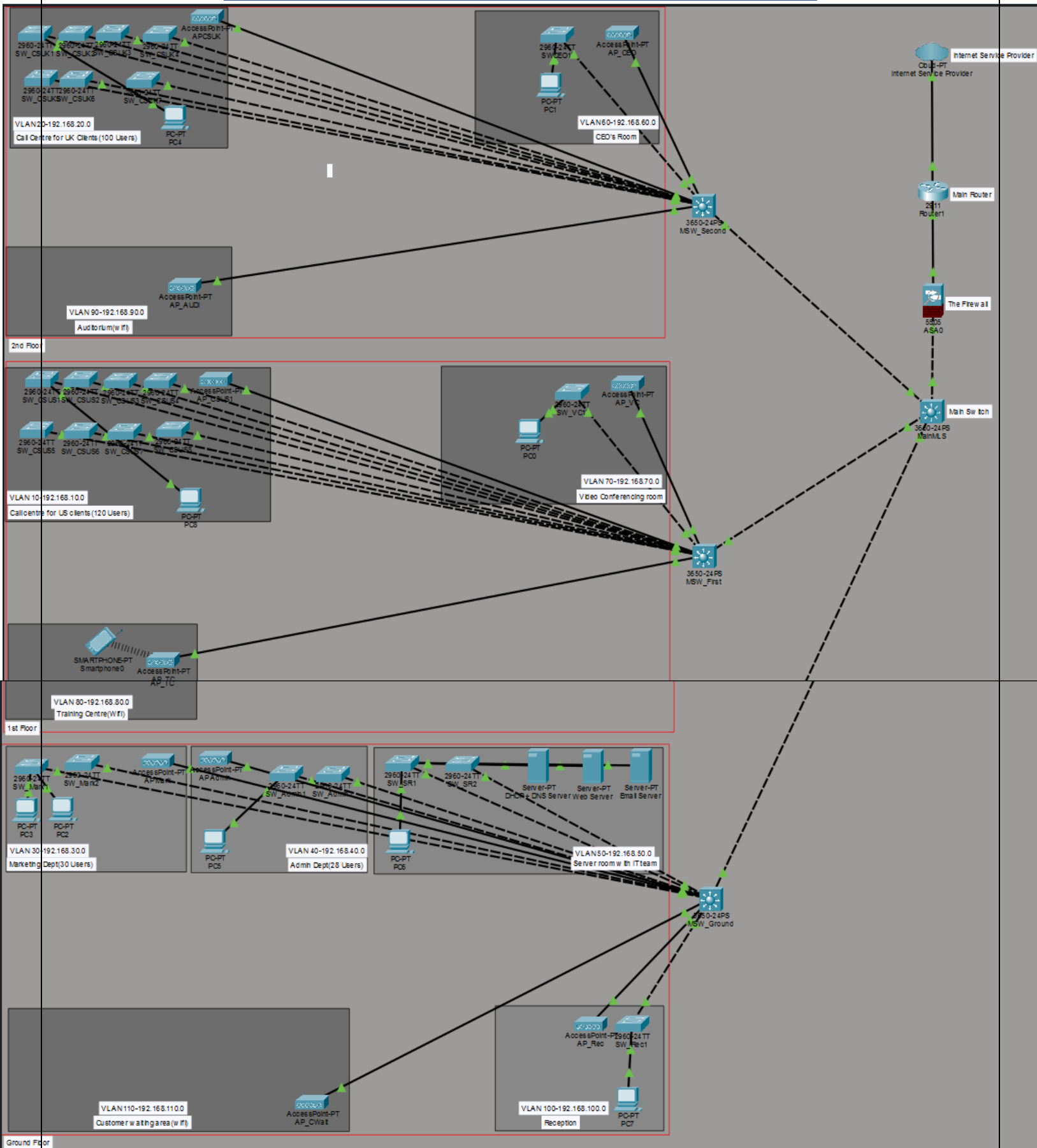
2nd floor,

- UK call center for 100 users
- Auditorium with wifi
- Ceos room

And the special requirements are,

- **There should be separate VLANs for departments**
- **All users should be able to use wifi**
- **Data traffic can increase by 50%**
- **Network should be fast and reliable**
- **All the devices are monitored**

Diagram I made for ConnectX using cisco packet tracer



Question 1

Networking devices ,servers and softwares I used for my network

❖ Switches

In my diagram as for the main and 3 floor switches I used the **cisco 3650-24PS switch** model switch,since it is a good reliable enterprise level switch which can be use as layer 2 and layer 3 switch so we dont need to use dedicated routers for each floor so it will save money.

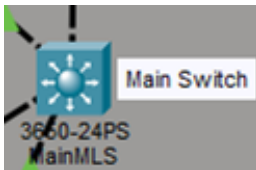


Figure 4:Main switch

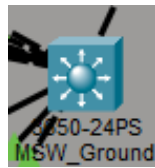


Figure 3:Ground floor switch

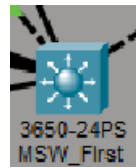


Figure 2:first floor switch

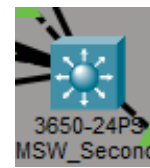


Figure 1:second floor switch

And for all departments I used the **cisco 2960 switch**.it is a layer 2 switch and supports VLAN,fast/gigabite ethernet.in my network i deployed 24 2960 switches as follows,

- 8 switches for US call center
- 7 switches for UK call center
- 2 switches for marketing dept
- 2 switches for admin dept
- 2 switches for server room
- 1 switch for reception
- 1 switch for video conferencing room
- 1 switch for ceos room



Figure 5:2960 switch

I used 2 switches for server room because,

- To improve security by keeping severs and it teams internet connection separately
- Speed up the internet performance
- If one switch fail,there is another switch as a backup

❖ Router

As the main router of my network I chose the **cisco 2911 router** and connected it to the internet via cloud pt.



Figure 6:Main router

my ports are,

GigabitEthernet0/0-connected to cloud(public ip-203.0.113.1)

GigabitEthernet0/1-connected to firewall(private ip-192.168.0.1)

❖ Firewall

I chose the **cisco ASA 5505 firewall** as my firewall and placed it between router and main switch for a better security.

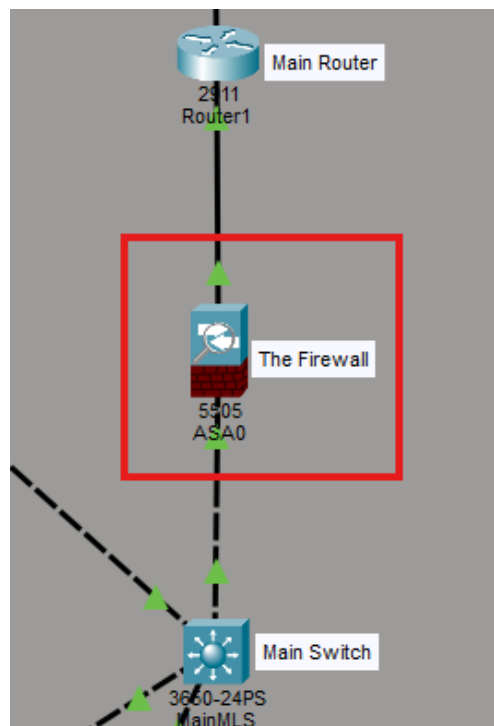


Figure 7:Firewall

❖ Access points

I deployed wireless Aps in every department so anyone can access wifi as the requirements and used **cisco AP-PT** for that.

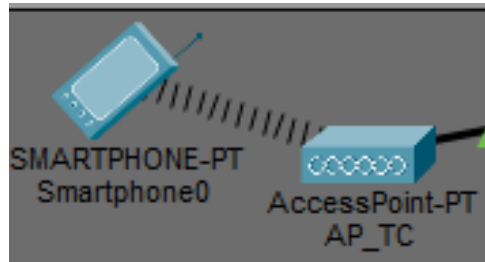


Figure 8:Access point

every AP, I configured with a unique SSID and a WPA2-PSK password for higher security.

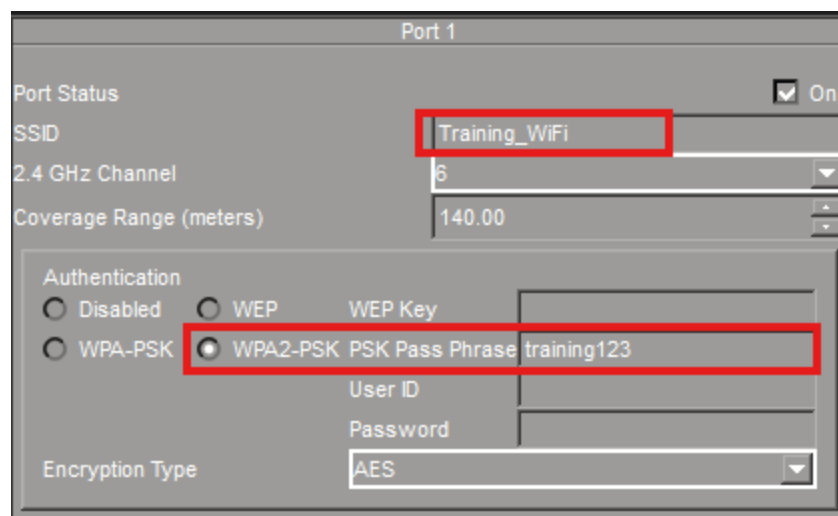


Figure 9:SSID & Password

❖ Servers

I used **cisco Server-PT** s and created three servers for my network on the ground floors server room.

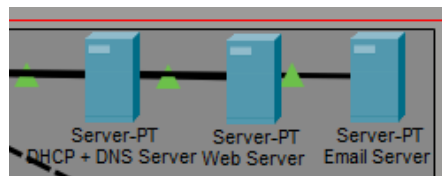


Figure 10:Servers

Server 1-for DHCP & DNS server(192.168.50.2)

Server 2-for WEB server(192.168.50.3)

Server 3-for Email server(192.168.50.4)

And tested them,

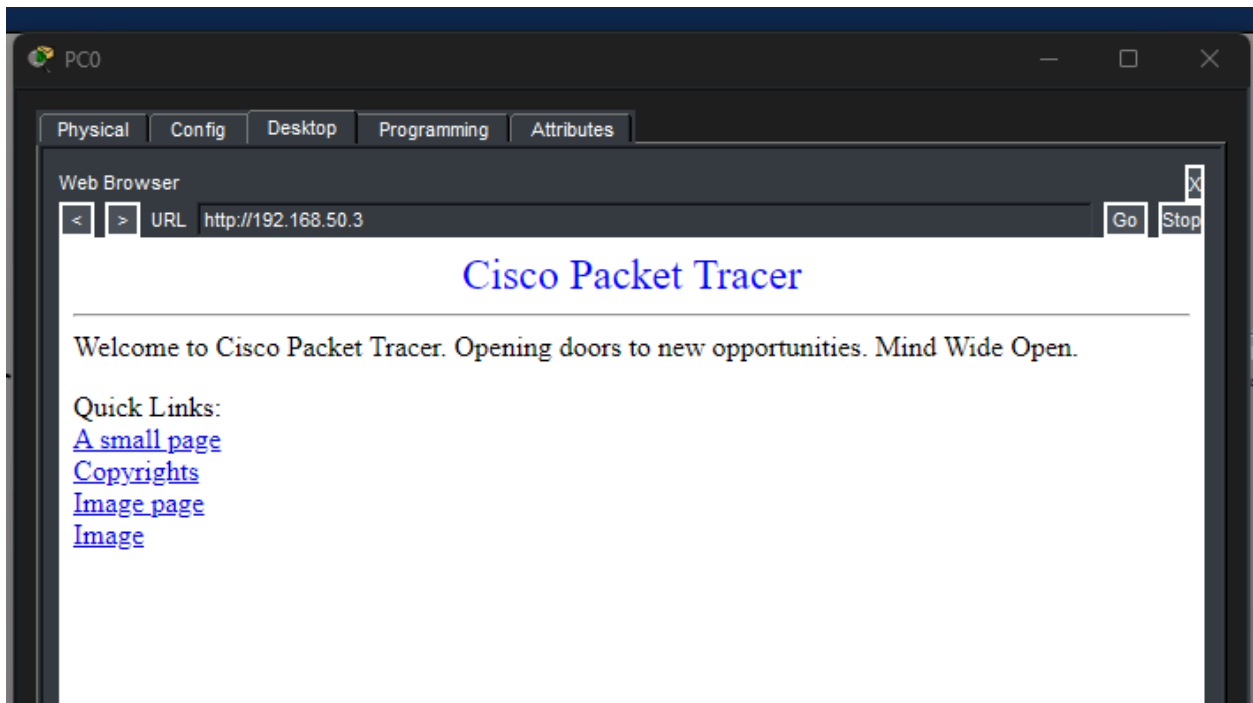


Figure 11: Servers in web browser

❖ DHCP configuration

I configured my DHCP on my main multilayer switch and created 11 DHCP pools. each pool has a network address ,default gateway. With DHCP any device that connects to my network will receive an ip address automatically.

```
MainMLS>enable
MainMLS#show ip dhcp pool

Pool VLAN10 :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 1
  Excluded addresses                : 11
  Pending event                     : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  192.168.10.1       192.168.10.1 - 192.168.10.254  1 / 11 / 254

Pool VLAN20 :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 1
  Excluded addresses                : 11
  Pending event                     : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  192.168.20.1       192.168.20.1 - 192.168.20.254  1 / 11 / 254
```

Figure 12: DHCP pools

❖ Softwares

These are the softwares I used for my network,

Software Name	Category	Used for
1. Windows 11 pro	OS	Installed to all end pcs in the network
2. Windows server 2024	OS	Installed to Servers
3. PRTG network monitor	Monitoring	Free to use,easy to setup,use to monitor the network
4. Microsoft defender for business	Virus guard	Installed to all pcs,built in &easy to use
5. Windows server backup	Backup	Installed to server pcs(free)



Figure 14:Windows 11 pro



Figure 13:Windows server



Figure 15:PRTG network monitor



Figure 17:Windows defender



Figure 16:Windows server backup

Question 2

Ip address allocation table

Department	VLAN	Oiriginal users	After 50%	Network address	Subnet mask	Gateway	Ip range
Call center US	10	120	180	192.168.10.0	255.255.255.0	192.168.10.1	.10.2— 10.254
Call center UK	20	100	150	192.168.20.0	255.255.255.0	192.168.20.1	.20.2— 20.254
Marketing	30	30	45	192.168.30.0	255.255.255.0	192.168.30.1	.30.2— 30.254
Admin dept	40	28	42	192.168.40.0	255.255.255.0	192.168.40.1	.40.2— 40.254
Server room	50	5	8	192.168.50.0	255.255.255.0	192.168.50.1	.50.2— 50.254
Ceos room	60	1	2	192.168.60.0	255.255.255.0	192.168.60.1	.60.2— 60.254
Video conference	70	2	3	192.168.70.0	255.255.255.0	192.168.70.1	.70.2— 70.254
Training center	80	50	75	192.168.80.0	255.255.255.0	192.168.80.1	.80.2— 80.254
Auditorium	90	50	75	192.168.90.0	255.255.255.0	192.168.90.1	.90.2— 90.254
Reception	100	2	3	192.168.100.0	255.255.255.0	192.168.100.1	.100.2— 100.254
Customer waiting	110	50	75	192.168.110.0	255.255.255.0	192.168.110.1	.110.2— 110.254

Assumptions I made while making ip addresses,

- As per requirements they already said about a 50% data traffic increase so incase in the future if traffic goes more than that,I allocated 254 usable ip addresses to each department.By doing that we will not have to redesign our network.
- I assumed that users may use multiple devices to connect to the network so ip address range should be large enough for that.
- By making separate VLANs I assumed that each VLAN needs a separate security level.

Question 3

Network topologies I used for my network

I designed my network using a 3 tier hierarchical model which is perfect model for a call center network.

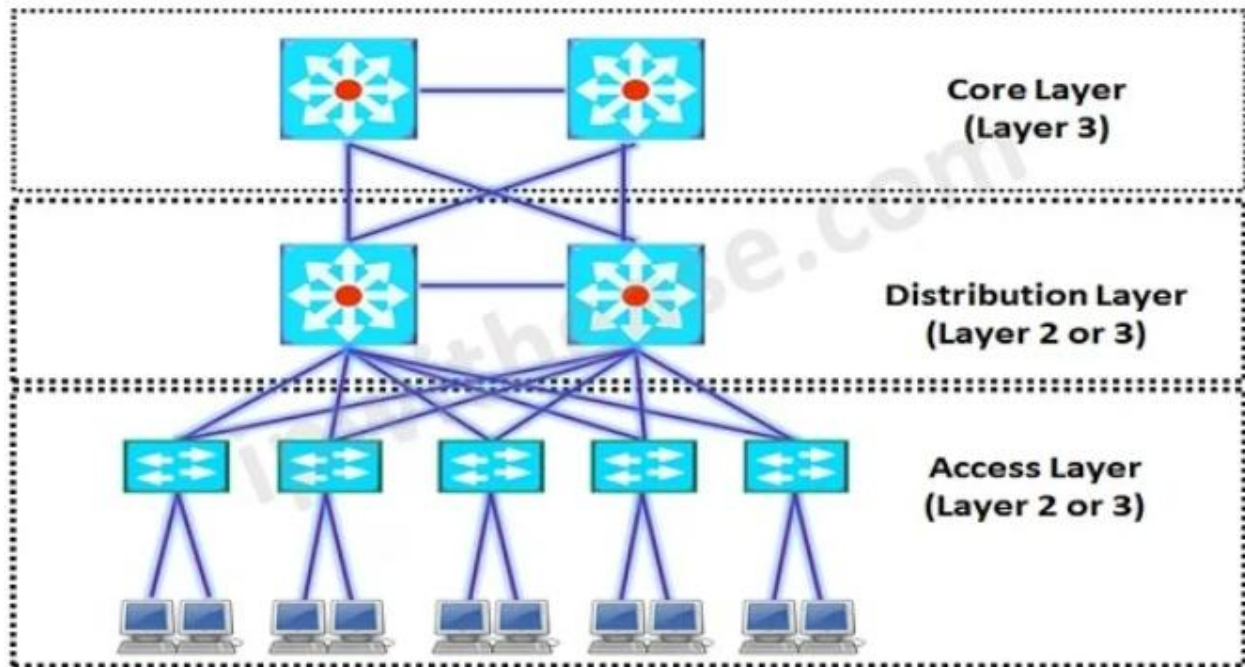
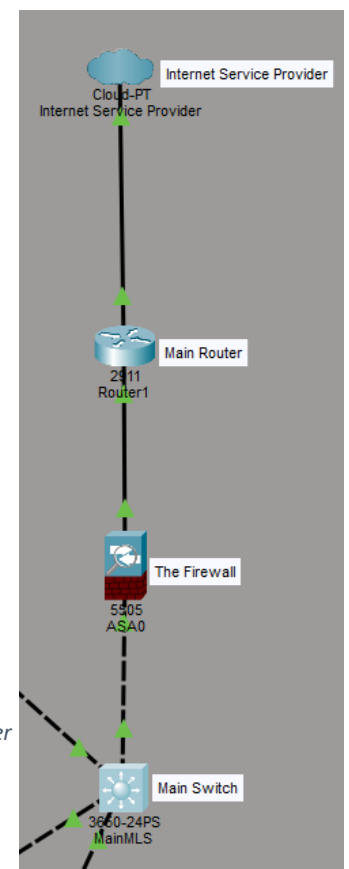


Figure 18:3 tier model

➤ so on my core layer,

I used the **point to point topology** till the main switch,

Figure 19:Core layer



- The Main switch distributed the connection to the 3 floor switches,
Using the **star topology**.

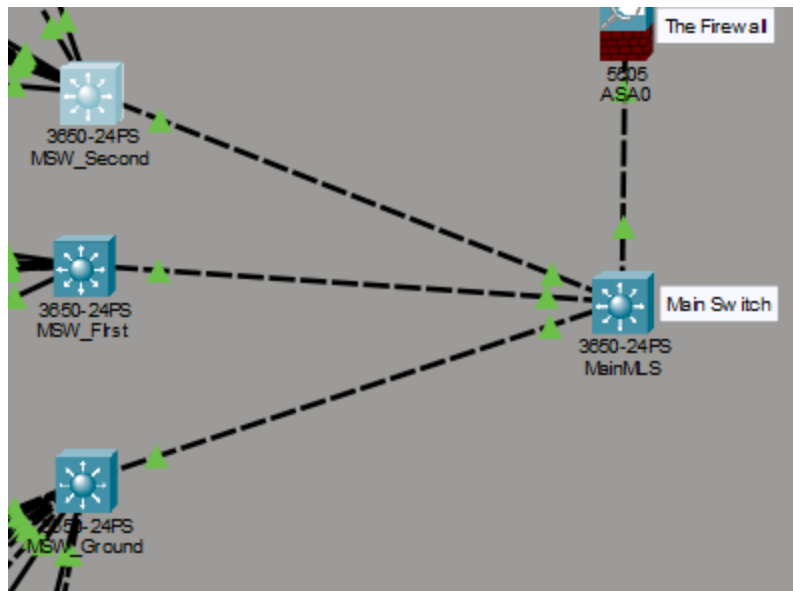


Figure 20:Main switch distribution

- Each floor switch distributed the connection to the departments,
Using the **star topology**.

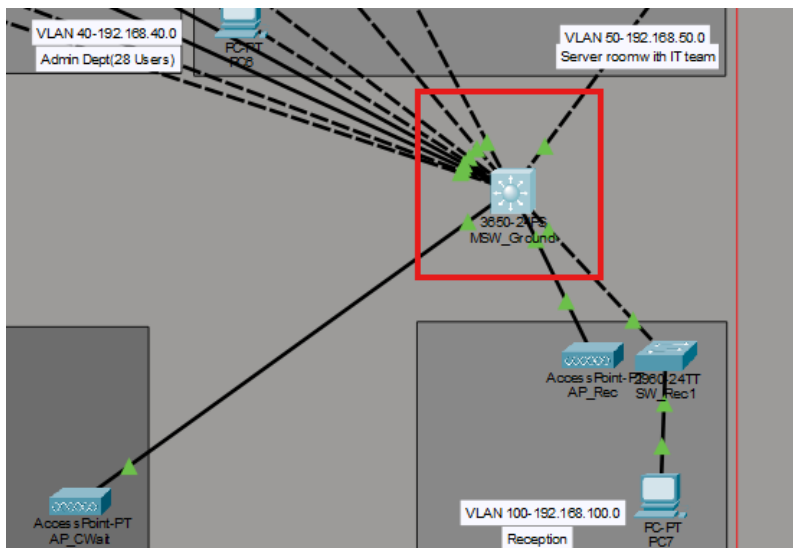


Figure 21:Floor level distribution

- Finally on the access layer the connection distributed to end pcs,
Using the **star topology**.

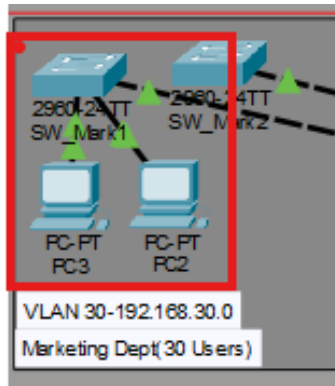


Figure 22:Access level distribution

So I created my network primarily based on the star topology because,

- To reduce the latency by direct connecting them
- To minimize failures
- To improve reliability
- Easy to maintain
- Easy to scale

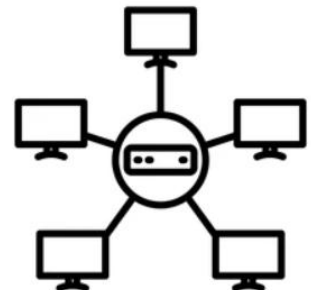


Figure 23:Star topology

Question 4

Recommended maintenance schedule

Schedule	Task	Maintained by
Daily	Check network monitoring logs	IT admin
Daily	Check servers status(online or not)	IT admin
Daily	Review firewall logs	Net admin
Weekly	Check switches and Aps logs	IT team
Weekly	Test backup and restorations	IT admin
Weekly	Check Ap connectivity status	IT support
Monthly	Update firmwares on network devices	Net admin
Monthly	Test VLAN routings	Net admin
Quarterly	Run virus scans	IT manager
Quarterly	Check full network performance	IT manager
Annually	Replace dead or bad hardware	IT manager
Annually	Check for new devices and scalability	IT manager

Question 5

Importance of network monitoring

In a call center like this network monitoring is a important thing we must do.specially in this network while handling multiple US and UK client calls if the connection gets loss,it will affect the business,customers and financial state as well.

With network monitoring tools we can continuously observe the network and its performance.

Benefits of using network monitoring tools,

- We can detect any failures early
- We can identify bandwidth bottlenecks
- We can detect security threats

In our network network monitoring is needed because we have larger number of devices.

So as I mentioned earlier I recommend **PRTG network monitor**

and here is why,

- Easy to setup
- Supports all cisco devices
- Can use real time monitoring

With PRTG network monitor we can monitor,

Device	What PRTG monitors
Main MLS (3650)	CPU, memory, port status, inter-VLAN traffic
Floor MLS switches	Port availability, bandwidth per floor
2960 Access switches	Port up/down status, connected devices
Router (2911)	WAN bandwidth, NAT connections
ASA Firewall	Connection attempts, blocked traffic
Access Points	WiFi client count, signal strength
Servers	CPU, memory, disk usage, service availability

Figure 24 monitoring table

Question 6

Backup strategy for my network

As a call center data is one of the most important parts in my network.while handling many calls any customers data loss will make a huge impact on business side.A good backup strategy can minimalize data loss.

So here is my recommended backup strategy,

❖ Im using 3-2-1 backup strategy which is

- ❖ 3 copies
- ❖ 2 different storages
- ❖ 1 offsite copy

And here is how I recommend to do that,

Full backup	Weekly every Sunday 2 a.m.	• A full backup of all the data • Doing this at night to minimize the impact of network performance • Stored on a server
Incremental backup	Daily every night 11 p.m.	• Backing up only the changed data since last backup • Same as doing this at night • if we cant access the full backup,we can use this worst case scenario is we loss only 24h data
Cloud backup	Daily every night 12 a.m.	• a daily backup which is going to store at a cloud storage to avoid physical disasters • we can access this backup from anywhere in the world

My justification for choosing this backup strategy,

This strategy provides all kinds of protections to all kinds of data loss scenarios.

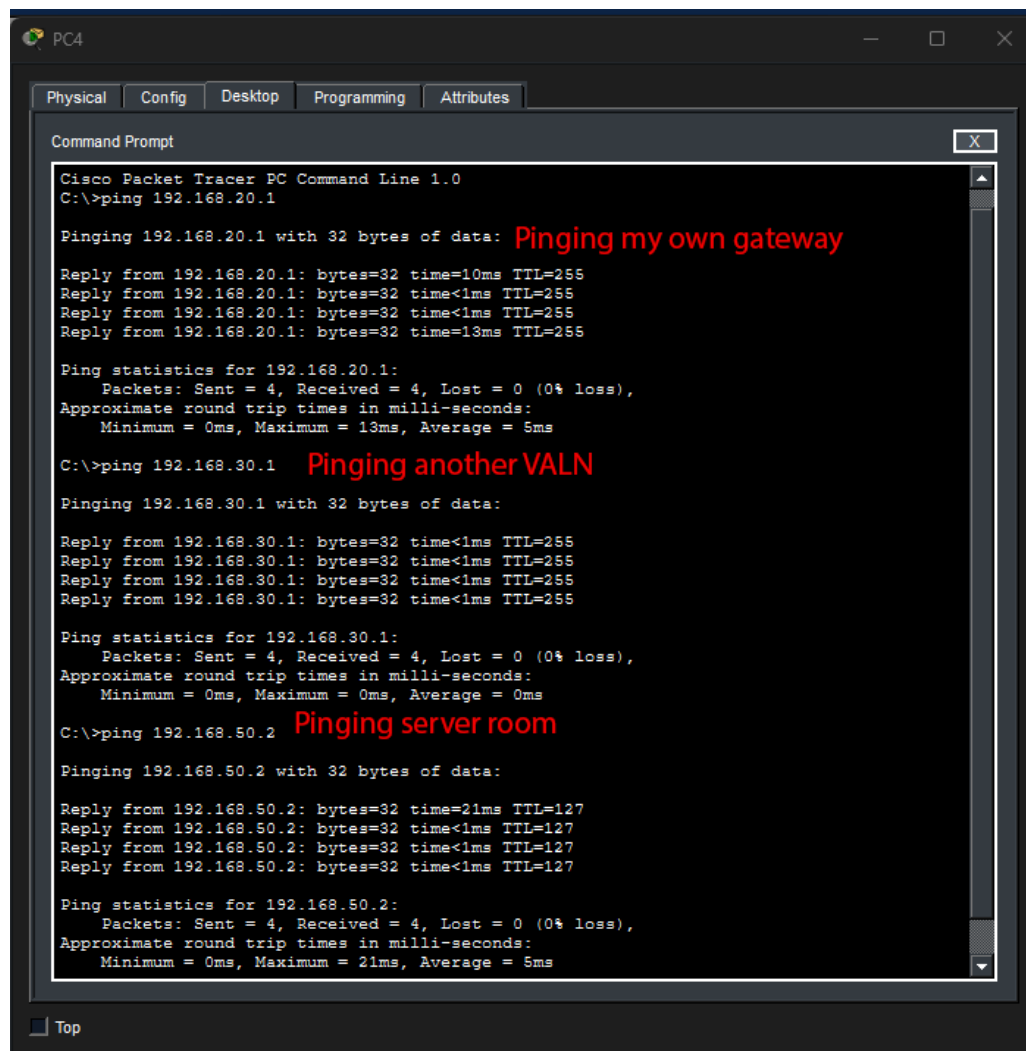
Full weekly backups protect all data and make it easy to recover it completely,

Daily incremental backups help to recover data with only a 24h data loss and

Cloud backups will protect data against physical disasters.

With all of these together it makes the 3 2 1 rule and give a high level of security to data in the call center.

Tests I did to make sure my network is working



```
PC4
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data: Pinging my own gateway

Reply from 192.168.20.1: bytes=32 time=10ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=13ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 5ms

C:\>ping 192.168.30.1 Pinging another VALN

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.50.2 Pinging server room

Pinging 192.168.50.2 with 32 bytes of data:

Reply from 192.168.50.2: bytes=32 time=21ms TTL=127
Reply from 192.168.50.2: bytes=32 time<1ms TTL=127
Reply from 192.168.50.2: bytes=32 time<1ms TTL=127
Reply from 192.168.50.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.50.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 21ms, Average = 5ms

Top
```

Figure 25 Pinging test

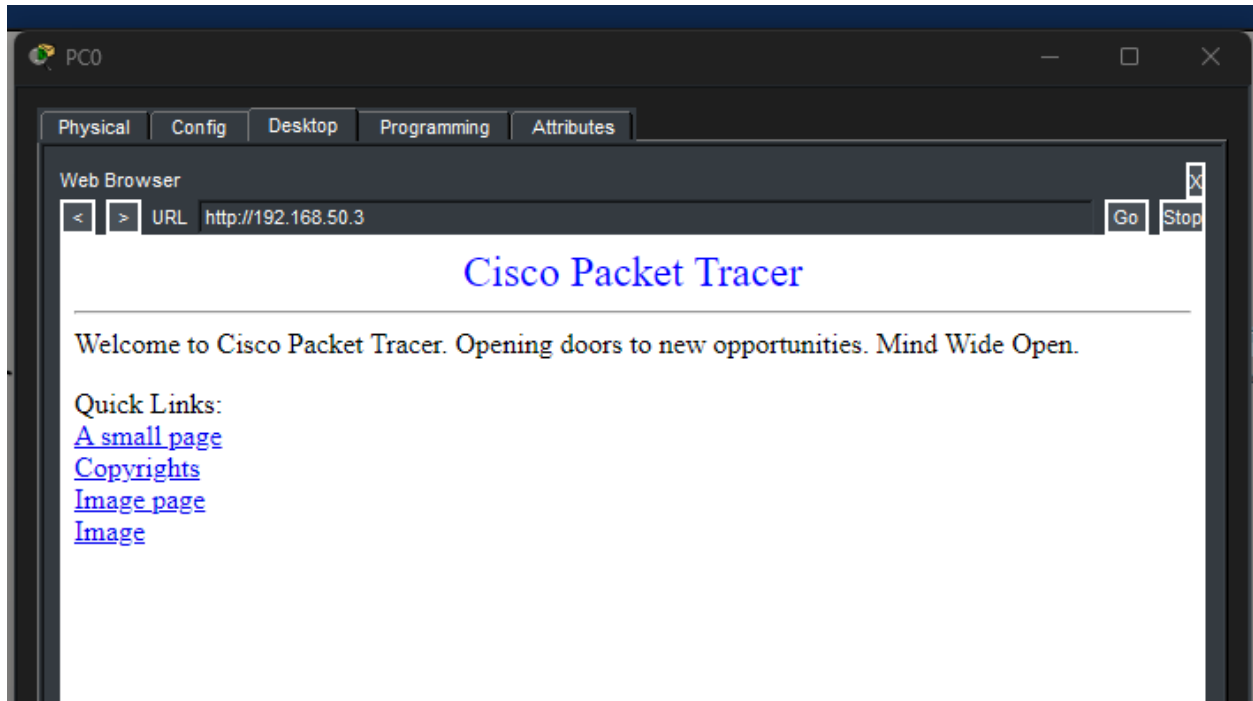


Figure 26 Servers testing

Conclusion

In conclusion, This report includes the network design for a call center named ConnectX and other ip address and routing details about it. This will ensure good communication, reliable and scalability in the network. Finally the maintenance methods and backup strategies as well.